

Bedanta Bandyopadhyay

AI-Driven ECE Undergrad | 3D Generalist | CAD & Simulation Specialist

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SUMMARY

I am a driven tech enthusiast currently pursuing a degree in Electronics and Communication Engineering. With a strong passion for 3D design, computer graphics, and programming, I focus on building innovative and impactful solutions. My skill set spans coding, CAD modeling, artificial intelligence, machine learning, deep learning, automotive design, and energy-efficient systems. As an active participant in technical events and conferences, and a department rank holder, I am committed to blending creativity with technology to deliver exceptional outcomes.

EXPERIENCE

Co-Leader and Freelance 3D Generalist

Various clients and Freelance

04/2019 - Present Remote

Freelance and professional work in 3D design and automotive rendering

- Co-Leader:** Co-leading a brand specializing in automotive design and rendering for over 2 years.
- Freelancer:** Delivered 3D designs and animations as a Freelance 3D Generalist for 30+ global clients for over six years, building a reputation for creativity and quality.
- 3D Generalist and Composition Artist: [Full-Time]** At The Autofocus Studio, India, working with multiple clients and expanding the studio's portfolio.

Internship

Intern

09/06/2025 - 03/07/2025 IEDC, IEM Newtown, Kolkata

Internship on advanced and cleaner materials in solar technology

- Conducted performance optimization of Tri-Layer Lead-free Carbon based Perovskite Solar Cells using Machine Learning.
- Crafted a full detailed report on the entire process.

EDUCATION

B.Tech in Electronics and Communication Engineering

Institute of Engineering and Management

GPA (4th Semester): 9.18/10

07/2023 - Present Newtown, Kolkata

LANGUAGE and CO-CURRICULAR

English ★★★★★ Fluent Bengali ★★★★★ Fluent Hindi ★★★★★ Fluent

Media and Tech Lead : Iris The Photographic Society of UEMK

11/2024 - 08/2025 Newtown, Kolkata

Graphics Lead : CSI UEMK Students' Chapter

11/2024 - Present Newtown, Kolkata

Manager : CSI UEMK Students' Chapter

08/2025 - Present Newtown, Kolkata

CERTIFICATIONS

Business Intelligence and Analytics : NPTEL

06/2025 IIT Madras

Business strategies for big data handling and analysis.

Joy of Computing with Python : NPTEL

11/2024 IIT Ropar

Basic to advanced level of Python programming.

Cloud Computing : NPTEL

06/2025 IIT Kharagpur

Introduction to Cloud Computing and moving up to advanced topics.

Introduction to Internet of Things : Coursera

10/2024 University of San Diego

Introduction to IoT and familiarisation with various tools.

STRENGTHS

Project Management and Communication

Skilled in Project Management with a strong ability to communicate.

Creative Problem Solving

Creative problem solving approaches with a strong focus on innovative design.

KEY ACHIEVEMENTS

Glaucoma Detection by Deep Learning (Ongoing)

Developed an AI-based Glaucoma Detection System using retinal imaging and deep learning, aimed at early diagnosis and accessible eye health screening via a user-friendly platform.

Domestic Charger for Bajaj Chetak EV

Designed a charging solution for Bajaj Chetak EV showcased at Kshitij-24 (IIT KGP TechFest).

Import Enhancement Add-on for Blender3D

Created an add-on converting ASCII FBX files into Binary FBX for Blender3D, enhancing workflow.

SIEM System

Engineered a security information and event management (SIEM) system for tracking user activity logs.

2-Switch Current-Controlled Buck-Boost Converter

Developed a power-efficient converter for higher output with minimal energy usage.

SKILLS

- Languages: Python, C, Java, JavaScript, Rust.
- AI-ML-DL: TensorFlow, Keras, Scikit-learn, PyTorch.
- Tools: Git
- Web: Flask, Node.js, Next.js, Express.js HTML, CSS
- Database: MongoDB
- Simulation: Cadence Virtuoso, LT Spice, SCAPS 1D, HFSS.
- 3D Modelling: 3DS Max, Blender, Maya.
- CAD Designing and Modeling: AutoCAD, Fusion360.
- Physical Render Engines: Cycles, Corona, Vray, Luxcore, Keyshot3D.
- Real Time Render Engines: Eevee, Unreal Engine.
- Composition: Blackmagic Design Fusion Studio.
- CG Tools: Adobe Suite, Corel Suite.

